

The Double Angle Identities

I) $\sin(A+B) = \sin A \cos B + \sin B \cos A$

$\sin(A+A) = \sin A \cos A + \sin A \cos A$

$\sin(2A) = 2 \sin A \cos A$

II) $\cos(A+B) = \cos A \cos B - \sin A \sin B$

$\cos(A+A) = \cos A \cos A - \sin A \sin A$

$\cos(2A) = \cos^2 A - \sin^2 A$

III) $\cos(2A) = \cos^2 A - \sin^2 A$ $\sin^2 A + \cos^2 A = 1$

$\cos(2A) = 1 - \sin^2 A - \sin^2 A$ $\cos^2 A = 1 - \sin^2 A$

$\cos(2A) = 1 - 2 \sin^2 A$

IV) $\cos(2A) = \cos^2 A - \sin^2 A$ $\sin^2 A + \cos^2 A = 1$

$\cos(2A) = \cos^2 A - (1 - \cos^2 A)$ $\sin^2 A = 1 - \cos^2 A$

$\cos(2A) = \cos^2 A - 1 + \cos^2 A$

$\cos(2A) = 2 \cos^2 A - 1$

Prove that the following equations are true by converting the left side to the right.

• ① $\frac{\sin(2\theta)}{\cos^2 \theta} = 2 \tan \theta$

$\frac{2 \sin \theta \cos \theta}{\cos^2 \theta} =$

$\frac{2 \sin \theta \cos \theta}{\cos \theta \cos \theta} =$

$\frac{2 \sin \theta}{\cos \theta} =$

$2 \tan \theta =$

③ $\frac{2}{\sin(2w)} = \csc w \sec w$

$\frac{2}{2 \sin w \cos w} =$

$\frac{1}{\sin w \cos w} =$

$\frac{1}{\sin w} \cdot \frac{1}{\cos w} =$

$\csc w \sec w =$

② $\cos(2A) + \sin^2 A = \cos^2 A$

$\cos^2 A - \sin^2 A + \sin^2 A =$

$\cos^2 A =$

④ $\frac{1 - \cos(2B)}{2} = \sin^2 B$

$\frac{1 - (1 - 2 \sin^2 B)}{2} =$

$\frac{1 - 1 + 2 \sin^2 B}{2} =$

$\frac{2 \sin^2 B}{2} =$

$\sin^2 B =$

$$\bullet \textcircled{5} \frac{2\sin^2\alpha}{\sin(2\alpha)} = \tan\alpha$$

$$\frac{2\sin^2\alpha}{2\sin\alpha\cos\alpha} =$$

$$\frac{2\sin\alpha\sin\alpha}{2\sin\alpha\cos\alpha} =$$

$$\frac{\sin\alpha}{\cos\alpha} =$$

$$\tan\alpha =$$



$$\textcircled{7} \frac{\sin\theta\cos\theta}{\sin(2\theta)} = \frac{1}{2}$$

$$\frac{\sin\theta\cos\theta}{2\sin\theta\cos\theta} =$$

$$\frac{1}{2} =$$



$$\textcircled{6} \frac{\cos(2\beta) + 1}{2} = \cos^2\beta$$

$$\frac{2\cos^2\beta - 1 + 1}{2} =$$

$$\frac{2\cos^2\beta}{2} =$$

$$\cos^2\beta =$$



$$\textcircled{8} \cos(2\beta) - \cos^2\beta = -\sin^2\beta$$

$$\cos^2\beta - \sin^2\beta - \cos^2\beta =$$

$$-\sin^2\beta =$$



$$\bullet \textcircled{9} \frac{\cos^2 w}{\sin(2w)} = \frac{\cot w}{2}$$

$$\frac{\cos^2 w}{2\sin w \cos w} =$$

$$\frac{\cos w}{2\sin w} =$$

$$\frac{1}{2} \cdot \frac{\cos w}{\sin w} =$$

$$\frac{1}{2} \cdot \cot w =$$

$$\frac{\cot w}{2} =$$



$$\textcircled{10} \frac{\cos(2\beta)}{\cos^2\beta} = 2 - \sec^2\beta$$

$$\frac{2\cos^2\beta - 1}{\cos^2\beta} =$$

$$\frac{2\cos^2\beta}{\cos^2\beta} - \frac{1}{\cos^2\beta} =$$

$$2 - \sec^2\beta =$$



$$\textcircled{11} \frac{1 - \sin \beta}{\sin(2\beta)} = \frac{\csc \beta \sec \beta - \sec \beta}{2}$$

$$\frac{1 - \sin \beta}{2 \sin \beta \cos \beta} =$$

$$\frac{1}{2 \sin \beta \cos \beta} - \frac{\sin \beta}{2 \sin \beta \cos \beta} =$$

$$\frac{1}{2 \sin \beta \cos \beta} - \frac{1}{2 \cos \beta} =$$

$$\frac{1}{2} \cdot \frac{1}{\sin \beta} \cdot \frac{1}{\cos \beta} - \frac{1}{2} \cdot \frac{1}{\cos \beta} =$$

$$\frac{1}{2} \cdot \csc \beta \sec \beta - \frac{1}{2} \sec \beta =$$

$$\frac{\csc \beta \sec \beta}{2} - \frac{\sec \beta}{2} =$$

$$\frac{\csc \beta \sec \beta - \sec \beta}{2} =$$

$$\textcircled{12} \frac{\cos(2A)}{\sin A + \cos A} = \cos A - \sin A$$

$$\frac{\cos^2 A - \sin^2 A}{\sin A + \cos A} = \cos A - \sin A$$

$$\frac{(\cos A + \sin A)(\cos A - \sin A)}{\sin A + \cos A} =$$

$$\cos A - \sin A =$$



$$\bullet \textcircled{13} \frac{\sin(2\theta)}{4 \sin^2 \theta} = \frac{\cot \theta}{2}$$

$$\frac{2 \sin \theta \cos \theta}{4 \sin^2 \theta} =$$

$$\frac{2 \cos \theta}{4 \sin \theta} =$$

$$\frac{2}{4} \cdot \frac{\cos \theta}{\sin \theta} =$$

$$\frac{1}{2} \cdot \cot \theta =$$

$$\frac{\cot \theta}{2} =$$



$$\textcircled{14} \frac{\cos(2\alpha)}{\sin^2 \alpha} = \csc^2 \alpha - 2$$

$$\frac{1 - 2 \sin^2 \alpha}{\sin^2 \alpha} =$$

$$\frac{1}{\sin^2 \alpha} - \frac{2 \sin^2 \alpha}{\sin^2 \alpha} =$$

$$\csc^2 \alpha - 2 =$$



$$(15) \frac{\cos B - 1}{\sin(2B)} = \frac{\csc B - \csc B \sec B}{2} \quad (16) \frac{\cos W - \sin W}{\cos(2W)} = \frac{1}{\cos W + \sin W}$$

$$\frac{\cos B - 1}{2 \sin B \cos B} =$$

$$\frac{\cos W - \sin W}{\cos^2 W - \sin^2 W} =$$

$$\frac{\cos B}{2 \sin B \cos B} - \frac{1}{2 \sin B \cos B} =$$

$$\frac{\cos W - \sin W}{(\cos W - \sin W)(\cos W + \sin W)} =$$

$$\frac{1}{2 \sin B} - \frac{1}{2 \sin B \cos B} =$$

$$\frac{1}{\cos W + \sin W} =$$

$$\frac{1}{2} \cdot \frac{1}{\sin B} - \frac{1}{2} \cdot \frac{1}{\sin B} \cdot \frac{1}{\cos B} =$$

$$\frac{1}{2} \csc B - \frac{1}{2} \csc B \sec B =$$

$$\frac{\csc B}{2} - \frac{\csc B \sec B}{2} =$$

$$\frac{\csc B - \csc B \sec B}{2} =$$

• (17) Solve if $0 < \beta < 360^\circ$

$$\sin^2 \beta + \cos(2\beta) = 0$$

$$\sin^2 \beta + \cos^2 \beta - \sin^2 \beta = 0$$

$$\cos^2 \beta = 0$$

$$\sqrt{\cos^2 \beta} = \pm \sqrt{0}$$

$$\cos \beta = \pm 0$$

$$\cos \beta = 0$$

$$\boxed{\beta = 90^\circ, 270^\circ}$$

(18) Solve if $0 \leq A \leq 360^\circ$

$$\sin(2A) = \sin A$$

$$2 \sin A \cos A = \sin A$$

$$2 \sin A \cos A - \sin A = 0$$

$$\sin A (2 \cos A - 1) = 0$$

$$\sin A = 0 \text{ or } 2 \cos A - 1 = 0$$

$$A = 0^\circ, 180^\circ, 360^\circ \text{ or } \cos A = \frac{1}{2}$$

$$A = 60^\circ, 300^\circ$$

$$\boxed{A = 0^\circ, 60^\circ, 180^\circ, 300^\circ, 360^\circ}$$

①9 Solve if $0 < \theta \leq 360^\circ$

$$1 - \cos(2\theta) = 3\cos^2\theta$$

$$1 - (2\cos^2\theta - 1) = 3\cos^2\theta$$

$$1 - 2\cos^2\theta + 1 = 3\cos^2\theta$$

$$2 - 2\cos^2\theta = 3\cos^2\theta$$

$$2 = 5\cos^2\theta$$

$$\frac{2}{5} = \cos^2\theta$$

$$\pm\sqrt{\frac{2}{5}} = \sqrt{\cos^2\theta}$$

$$\pm\frac{\sqrt{10}}{5} = \cos\theta$$

$$\cos\theta = \frac{\sqrt{10}}{5} \text{ or } \cos\theta = -\frac{\sqrt{10}}{5}$$

$$\theta = \cos^{-1}\left(\frac{\sqrt{10}}{5}\right) \text{ or } \theta = \cos^{-1}\left(-\frac{\sqrt{10}}{5}\right)$$

$$\theta \approx 50.77^\circ, 309.23^\circ \text{ or } \theta \approx 129.23^\circ, 230.77^\circ$$

②0 Solve if $0 < w < 360^\circ$

$$2\sin^2w - \sin(2w) = 0$$

$$2\sin^2w - 2\sin w \cos w = 0$$

$$2\sin w (\sin w - \cos w) = 0$$

$$2\sin w = 0 \text{ or } \sin w - \cos w = 0$$

$$\sin w = 0 \text{ or } \sin w = \cos w$$

$$w = 0^\circ, 180^\circ, 360^\circ \text{ or } \frac{\sin w}{\cos w} = \frac{\cos w}{\cos w}$$

$$\tan w = 1$$

$$w = 45^\circ, 225^\circ$$

$$w = 45^\circ, 180^\circ, 225^\circ$$

• ②1 Solve if $0 \leq B < 360^\circ$

$$\cos(2B) + \cos^2B = 0$$

$$2\cos^2B - 1 + \cos^2B = 0$$

$$3\cos^2B - 1 = 0$$

$$3\cos^2B = 1$$

$$\cos^2B = \frac{1}{3}$$

$$\sqrt{\cos^2B} = \pm\sqrt{\frac{1}{3}}$$

$$\cos B = \pm\frac{\sqrt{3}}{3}$$

$$\cos B = \frac{\sqrt{3}}{3} \text{ or } \cos B = -\frac{\sqrt{3}}{3}$$

$$B = \cos^{-1}\left(\frac{\sqrt{3}}{3}\right) \text{ or } B = \cos^{-1}\left(-\frac{\sqrt{3}}{3}\right)$$

$$B = 54.74^\circ, 305.26^\circ \text{ or } B = 125.26^\circ, 234.74^\circ$$

②2 Solve if $0 \leq w < 360^\circ$

$$2\cos w - \sin(2w) = 0$$

$$2\cos w - 2\sin w \cos w = 0$$

$$2\cos w (1 - \sin w) = 0$$

$$2\cos w = 0 \text{ or } 1 - \sin w = 0$$

$$\cos w = 0$$

$$w = 90^\circ, 270^\circ \text{ or } 1 = \sin w$$

$$w = 90^\circ$$

$$w = 90^\circ, 270^\circ$$

②③ Solve if $0 \leq \alpha \leq 360^\circ$ | ②④ Solve if $0 < \theta \leq 360^\circ$

$$\cos(2\alpha) - 1 = -1 + 2\sin^2\alpha$$

$$1 - 2\sin^2\alpha - 1 = -1 + 2\sin^2\alpha$$

$$-2\sin^2\alpha = -1 + 2\sin^2\alpha$$

$$-4\sin^2\alpha = -1$$

$$\sin^2\alpha = \frac{1}{4}$$

$$\sqrt{\sin^2\alpha} = \pm \sqrt{\frac{1}{4}}$$

$$\sin\alpha = \pm \frac{1}{2}$$

$$\sin\alpha = \frac{1}{2} \quad \text{or} \quad \sin\alpha = -\frac{1}{2}$$

$$\boxed{\alpha = 30^\circ, 150^\circ \text{ or } \alpha = 210^\circ, 330^\circ}$$

$$\sin(2\theta) = -2\cos^2\theta$$

$$2\sin\theta\cos\theta = -2\cos^2\theta$$

$$2\sin\theta\cos\theta + 2\cos^2\theta = 0$$

$$2\cos\theta(\sin\theta + \cos\theta) = 0$$

$$2\cos\theta = 0 \quad \text{or} \quad \sin\theta + \cos\theta = 0$$

$$\cos\theta = 0 \quad \text{or} \quad \sin\theta = -\cos\theta$$

$$\theta = 90^\circ, 270^\circ \quad \text{or} \quad \frac{\sin\theta}{\cos\theta} = -\frac{\cos\theta}{\cos\theta}$$

$$\text{or } \tan\theta = -1$$

$$\text{or } \theta = 135^\circ, 315^\circ$$

$$\boxed{\theta = 90^\circ, 135^\circ, 270^\circ, 315^\circ}$$

• ②⑤ Solve if $0 < x < 360^\circ$

$$\cos(2x) - 3\sin x = 3\sin^2 x + 9\sin x + 5$$

$$1 - 2\sin^2 x - 3\sin x = 3\sin^2 x + 9\sin x + 5$$

$$0 = 5\sin^2 x + 12\sin x + 4$$

$$\text{Let } y = \sin x$$

$$0 = 5y^2 + 12y + 4$$

$$0 = (5y + 2)(y + 2)$$

$$5y + 2 = 0 \quad \text{or} \quad y + 2 = 0$$

$$y = -\frac{2}{5} \quad \text{or} \quad y = -2$$

$$\sin x = -\frac{2}{5} \quad \text{or} \quad \sin x = -2$$

$$x = \sin^{-1}\left(-\frac{2}{5}\right)$$

$$x \approx -23.58^\circ$$

$$\boxed{x \approx 203.58^\circ, 336.42^\circ}$$

- (26) Solve if $0 \leq \theta \leq 360^\circ$.

$$\cos \theta + \cos(2\theta) = 0$$

$$\cos \theta + 2\cos^2 \theta - 1 = 0$$

$$2\cos^2 \theta + \cos \theta - 1 = 0$$

$$\text{Let } x = \cos \theta$$

$$2x^2 + x - 1 = 0$$

$$(2x-1)(x+1) = 0$$

$$\swarrow \quad \searrow$$
$$2x-1=0 \quad \text{or} \quad x+1=0$$

$$x = \frac{1}{2} \quad \text{or} \quad x = -1$$

$$\cos \theta = \frac{1}{2} \quad \text{or} \quad \cos \theta = -1$$

$$\theta = 60^\circ, 300^\circ \quad \text{or} \quad \theta = 180^\circ$$

$$\boxed{\theta = 60^\circ, 180^\circ, 300^\circ}$$